

# Answers to Referees' Questions

## “Interpolating the Stochastic Heat and Wave Equations with Time-independent Noise: Solvability and Exact Asymptotics”

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We would like to thank the two anonymous reviewers for carefully reading our manuscript and for providing detailed reports, which have helped us to improve significantly the paper. We have revised our paper accordingly. Here are a list of changes that we have made. The page and equation numbers refer to the revised version of the manuscript.

The page and line numbers below refer to those from the original submission. In particular, the line number refers to the one added by the system at the left margin of the paper.

### 1 Changes according to Reviewer 1

1. page 2: We have added “-definite” in the phrase: “nonnegative and nonnegative-definite”.
2. page 4: Regarding more details of  $\mathcal{M}_{a,d}(f)$ , we have included one sentence to point out it is well defined. We recalled this quantity in Section 3.2.
3. page 13: We have switched the notation from  $\mathcal{A}_d$  to  $\mathcal{A}_{a,d}$  to be more precise.
4. page 15: We have included one footnote on p. 16 to give some references about the Wiens chaos expansion.
5. page 21: All  $\sigma(d, \alpha)$  in the proof of Lemmas 3.10 and 3.11 were replaced with  $\sigma(a, d, \alpha)$ .
6. page 29: “Now are are” has been updated to “Now we are”.

## 2 Changes according to Reviewer 2

1. page 1, line 21, 23 and 24: Changed “exits” to “exists”.
2. page 1, line 22: Inserted “as that” to the sentence.
3. page 2, line 32: Added parenthesis to  $m$  in the expression  $f^{(m)}$ .
4. page 2: We removed the mention of  $Z$  and  $Z^*$  and then rewrote the last paragraph to accommodate this change.
5. page 2, line 53: This is a good observation. Indeed, by Theorem 4.1 of [6] (originally [5]), when computing the homogeneous solutions, one only needs to use  $Z$  and  $Z^*$ , both of which do not depend on  $r$  as pointed out by the referee. The parameter  $r$  only appears in  $Y$ , which is used to compute the contribution by the driving force term.
6. page 4, line 48: Corrected the “Wiener” misspelling.
7. page 5 in the statement of Theorem 1.6: We corrected the condition on  $r$  from  $r \in [0, 1/2]$  to  $r \in [0, 1/2)$ . A white space has been added in part (2-i).
8. page 6, line 15: We added a space after “ $p \geq 2, \dots$ ”.
9. page 6, line 23: Added the word “illustrate”.
10. page 6, line 34: Changed “exits” to “exists”.
11. page 7, item[12.] (b): We referenced the theorem and figure below.
13. page 7, line 7: Changed “to” to “with”.
14. page 7, line 14: We removed the word “the”.
15. page 7, line 37 and line 48: Changed “exits” to “exists”.
16. page 7, line 41: Rewrote the paragraph starting on this line.
17. page 10, line 44: Changed “Laplace” to “Laplacian”.
18. page 10: We removed the repeated  $\alpha = a = d = 2$  case from the bottom of the page.
19. page 13, line 27: Added that  $\alpha < a$  as well.
20. page 13: Fixed the  $\gamma(x - y)$  typo.

21. page 14: Example 2.10 is a continuation of Example 2.4, where other papers were cited. Similarly, Example 2.11 is a continuation of Example 2.5 with references given there. We try to obtain formulas as explicit as possible. In these two examples, we can make our formulas completely explicit when working under specific cases. Even though there were no previous asymptotic formulas in these two cases to compare with, we hope that the explicit formulas themselves are interesting enough to be presented.
22. page 14, line 3: Changed “exits” to “exists”.
23. page 15, line 11: Added a reference to the Skorohod integral at the end of this paragraph.
24. page 15, line 18: We have added additional information about  $G$  on page 2 to avoid confusion; see Eq. (1.2).
25. page 15, line 31: Reworded the sentence to say that the moment “diverges to infinity” for  $t > T_2$ .
26. page 15, line 39: Fixed the misspelling for “Wiener”.
27. page 16, line 18: The same to Item 24 above.
28. page 16, line 20: Fixed the order of the arguments for the kernel  $\tilde{f}_n$ .
29. page 16, line 38: Fixed the “Wiener” misspelling.
30. page 16, line 40: Changed  $R^d$  to  $\mathbb{R}^d$ .
31. page 17, line 23: Added the assumption that  $\alpha < \min\{2a, d\}$ .
32. page 17, Theorem 3.5: Added the assumption that  $\alpha < \min\{2a, d\}$ .
33. page 19, Lemma 3.7: [Nick, can you make a comment on this question?](#)
34. page 20: Replaced “ $f$ ” with “ $b$ ” in the definition for  $\|\cdot\|_{q,w}$ .
35. page 20, line 23: Removed the reference to [1, Lemma 1.6].
36. page 21, line 9: Added a reference for the comparison.
37. page 21, Lemma 3.9: Added the assumption on  $\alpha$ . Directed the reader to the definition of  $\sigma(a, d, \alpha)$ . Replaced “2” with “ $\alpha$ ” in the exponent.
38. page 21, the statement of Lemma 3.10: [The proof of the lemma shows that the constant depends on the noise through the scaling property property  \$\gamma\$ ; hence it depends on  \$\alpha\$  \(not individual  \$\alpha\_i\$ \).](#)
39. page 21, Proof of Lemma 3.10: Made clear that Lemma 3.5 is from the current paper. Fixed the  $g$  typo inside the definition of  $\Lambda$ .

40. page 22, line 23: Replaced “ $\mathbb{R}$ ” with “ $\mathbb{R}_+$ ”.
41. page 22, lines 26 and 33: Replaced “2” with “ $\alpha$ ” in the exponent.
42. page 22: All  $\sigma(d, \alpha)$  were replaced with  $\sigma(a, d, \alpha)$ .
43. page 22: We reverted back to using  $\gamma$  instead of the Riesz kernel. ie equation (3.34).
44. page 22, Lemma 3.11: Replaced  $\gamma(x - y)$  with  $\gamma(x + y)$ . **The second last centered equations in the proof has been updated. (Nick, please explain these changes...)**
45. page 23, line 4: Corrected the centered equation.
46. page 23: Reverted back to using  $\gamma(\cdot)$  instead of  $|\cdot|^{-\alpha}$ .
47. page 23, line 48: Replaced “ $f$ ” with “ $g$ ”.
48. page 24, line 3: Added reference to Lemma 3.11 for the use of  $\sigma$ . **Nick, you didn’t answer the concern of the referee: is the  $\sigma(a, d, \alpha)$  in Lemma 3.11 the same as that in Lemma 3.10?**
49. page 24: Removed the excess periods, fixed the exponents by changing “2” to “ $\alpha$ ”, and also replaced “ $\sigma(d, \alpha)$ ” with “ $\sigma(a, d, \alpha)$ ”.
50. page 25, line 3: Made it clear we are going to prove part (1) of Theorem 1.6.
51. page 25, Eq 4.5: Fixed the  $\xi_{\sigma(j)}$  error.
52. page 25, line 50: Gave a reference to the function  $K$ .
53. page 26, Eq 4.6: **The notation  $\mu(\vec{\xi})$  have explained at its first appearance — after (3.15).**
54. page 26 and 27, Eq 4.2:  $\alpha < 2a$  was assumed in the statement of the theorem and we added a reference to clarify that. We also reordered the argument as suggested.
55. page 27, Eq 4.14: Replaced “ $n$ ” with “2”.
56. page 27, line 28: Added a reference for the inequality, which is indeed true under Eq. (4.16).
57. page 27, line 35: Made a comment about the switch to  $x = 0$ .
58. page 28, Lemma 5.1: Relaxed the assumption on  $\alpha$  and referenced  $\rho$  as was suggested.
59. page 28, line 59: Added a reference for  $H_n$ .
60. page 29: Fixed all the +1 typos.
61. page 29: Explained the derivation of  $C_n$  – an application of Stirling’s formula in the form of Eq. (5.2).

62. page 29, line 56: Fixed the misspelling for the word “hypercontractivity”.
63. page 30: You were correct, we made the change to  $r$  ie  $r \in [0, 1/2)$ .
64. page 30, line 32: Fixed the typo involving the reference to Lemma 5.1.
65. page 32, line 37: Added “it”.
66. page 33, line 22: Fixed the subscript in  $\mathbb{E}(U_n(t))$ .
67. page 33, line 50: Replaced  $H$  with  $\mathcal{H}$ .
68. page 34, line 46 [Unsure what to change and the the  \$\rho\(g\)\$  comment.](#)
69. page 34, line 46: [Added that  \$\mathcal{F}^{-1}g \in \mathcal{H}\_+\$  which in return implies that  \$g \in L^2\(\mu\(dx\), \mathbb{R}^d\)\$](#)
70. page 34, line 49: [We have altered the argument to bound from below the quantity  \$\rho\(g\)\$  by they same thing but with the sup only over nonnegative  \$h\$ . In addition, we replaced  \$\rho\(g\)\$  with  \$\rho'\(g\)\$  so there is no confusion with the previous  \$\rho\$  defined in the paper.](#)
71. page 35, line 13: [Unsure what to say about the comment on  \$\mathcal{F}g\_h \* \widetilde{\mathcal{F}g\_h}\$ .](#)
72. page 35: Fixed all  $g_h$  typos.
73. page 37, Eq 6.19: Fixed the  $n_t$  typo.
74. page 38, line 10: Fixed the  $t_p$  typo.
75. page 39: Reformatted the proof of (4.9) and took into account the suggestions made.
76. page 40: Removed the duplicated reference.

### 3 Additional minor corrections

In addition to the above changes suggested by the two referees, we implemented some minor corrections which are highlighted in [blue](#) in the manuscript. We list these below.

1. page 31, Lemma 5.2: In the proof starting with the line “We want to find  $\beta$  and  $A$  so that...” we replaced  $R_n$  with  $R^n$  in the following centered equation.
2. Page 35, Prop 6.4: In the proof, we removed the line that calculated  $\|g\|_{W^{1,a}(\mathbb{R}^d)}$  as it was unneeded and also not correct.